Chapter 9 Earth in space

EXTRA WORKSHEET 2: Martians from Mars

VERBAL/LINGUISTIC **SCIENCE UNDERSTANDING**

Name:_____Class:_____Date: _____

In 1858, the Italian astronomer Angelo Secchi (1818–1878) reported seeing channels on the surface of Mars. These observations were confirmed in 1877 by another Italian astronomer, Giovanni Schiaparelli (1835–1910). Although neither astronomer gave an explanation as to what they were, American astronomer Percival Lowell (1855–1916) wrote three books (in 1895, 1906 and 1908) that suggested that the channels were canals dug by intelligent beings (Martians) to move water from the polar caps to irrigate their farms.

Many astronomers, however, couldn't see the vast network of canals that Lowell suggested and the channels that could be seen were far too wide to be dug by anyone. Although the 1910 *Encyclopaedia Britannica* expressed serious doubts about them, public imagination was ignited: intelligent Martians lived (or had lived) on Mars!

Lowell's canals were not disproved until 1965 when the *Mariner 4* space orbiter sent back the first close-up images of the surface of Mars. Some channels have been found to be natural canyons while others were an optical illusion created by lines of craters.

1 Two early astronomers reported seeing canals on the surface of Mars. **State**:

a their names

_____

b where they came from.

_____

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2 **Predict** whether they saw the canals with their naked eyes or with a telescope. **Justify** your answer.

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3 A twentieth century astronomer ‘confirmed’ the existence of canals on Mars.

a **Name** the astronomer

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b **State** his nationality

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c **State** his years of birth and death.

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d **Explain** what he thought the canals were used for.

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4 **Explain** why the existence of canals would suggest the existence of intelligent Martians.

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5 Many scientists in the early twentieth century doubted that the canals existed. **List** the evidence that cast doubt on the existence of natural canals on Mars.

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6 **Name** the publication that officially cast doubt on the canal theory.

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7 **Describe** the evidence that finally showed that the canals did not exist.

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8 Suppose intelligent life was discovered on Mars. **Propose** how humans on Earth would respond.

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